

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
IRNA Modification (GO:0006400)	0.19176247	66	7.358e-08	3.965e-04	METTL2B:155 THADA:190 PUS10:237 TRMT10A:259 DTWD1:305 FTSJ1:343
IRNA Methylation (GO:0030488)	0.24706383	36	2.936e-07	7.909e-04	METTL2B:155 THADA:190 TRMT10B:258 TRMT10A:259 FTSJ1:343 METTL8:374
Fatty Acid Catabolic Process (GO:0009062)	0.19125506	58	4.810e-07	8.639e-04	ECI2:10 ACAD11:177 SLC27A4:245 EHHAADH:348 ACOT8:398 HACL1:516
Chemical Synaptic Transmission (GO:00072	-0.09064213	254	7.310e-07	9.846e-04	NPTX1:32 RIMBP2:42 GABRR1:52 RPS6KA2:67 NTSR1:129 GABRB3:145
Muscle Contraction (GO:0006936)	-0.15221231	87	9.545e-07	1.029e-03	TACR2:22 MYH2:100 KCNJ12:182 LMDD2:252 MYH8:261 MYH1:341
DNA Repair (GO:0006281)	0.07865785	273	8.488e-06	4.739e-03	FANCM:50 FANCE:108 RHNO1:112 MBD4:124 MSH3:225 PARP4:284
RNA Methylation (GO:0001510)	0.17159860	57	7.590e-06	4.739e-03	METTL4:115 METTL2B:155 THADA:190 TRMT10A:259 FTSJ1:343 METTL8:374
RNA Modification (GO:0009451)	0.19039804	47	6.391e-06	4.739e-03	METTL2B:155 DTWD1:305 FTSJ1:343 TRMT1:713 PUS3:834 PUS7L:954
Regulation Of Synaptic Transmission, Glu	-0.17499561	54	8.795e-06	4.739e-03	TSZH3:95 TPRG1L:204 GRM6:268 DGK1:536 SHANK3:638 DRD2:654
Visual Perception (GO:0007601)	-0.13421604	95	6.356e-06	4.739e-03	TRPM1:17 VAX2:19 GNAT2:25 CRYAA:118 CABP2:122 OAT:133
Calcium Ion Import Across Plasma Membran	-0.22244737	33	9.842e-06	4.821e-03	TRPM1:17 SCN1A:219 CACNA1B:265 SCN2A:338 CLU:761 SLC8A3:855
Double-Strand Break Repair (GO:0006302)	0.10187575	157	1.113e-05	4.999e-03	HELQ:313 MCM9:315 TP53BP1:342 RNF168:352 DNA2:380 AUNIP:570
Striated Muscle Contraction (GO:0006941)	-0.17272792	52	1.665e-05	6.903e-03	MYH7B:132 MYH8:261 PGAM2:355 RYR2:386 TNNT3:391 KCNQ1:423
Double-Strand Break Repair Via Homologou	0.12081326	105	1.953e-05	7.014e-03	FANCM:50 HELQ:313 MCM9:315 XRCC1:337 AUNIP:570 ERCC5:638
Fatty Acid Beta-Oxidation (GO:0006635)	0.18069223	47	1.844e-05	7.014e-03	ECI2:10 ACOX2:63 ACAD11:177 GCDH:261 EHHAADH:348 ACOT8:398
Regulation Of Chromosome Separation (GO:	0.40811282	9	2.240e-05	7.543e-03	NCAPD2:110 SMC4:118 SMC2:883 NUMA1:1002 NCAPG:1278 TPR:1904
Modulation Of Chemical Synaptic Transmis	-0.11320509	117	2.424e-05	7.684e-03	NPTX1:32 TPRG1L:204 CLSTN1:256 CACNA1B:265 GRM6:268 CPLX3:408
IRNA Processing (GO:0008033)	0.17363651	49	2.645e-05	9.198e-03	METTL2B:155 DTWD1:305 FTSJ1:343 ADAT1:705 TRMT1:713 PUS3:834
Mitochondrial Gene Expression (GO:014005	0.12019508	100	3.371e-05	9.560e-03	PTCD3:246 LARS2:358 FASTKD1:446 MRPS10:475 TFB2M:498 MRPS9:696
Mitochondrial Translation (GO:0032543)	0.12089840	96	4.370e-05	1.177e-02	PTCD3:246 MTIF2:317 LARS2:358 MRPS10:475 TSFM:606 GFMI:695
Fatty Acid Oxidation (GO:0019395)	0.16547344	47	8.773e-05	2.251e-02	ECI2:10 ACAD11:177 EHHAADH:348 HACL1:516 ACADM:530 HACD2:573
Cell Junction Assembly (GO:0034329)	-0.12458697	82	9.814e-05	2.403e-02	SHANK2:13 DBNL8:81 SDK1:85 CDH26:277 CDH6:373 ITGA6:405
Regulation Of Heart Rate By Cardiac Cond	-0.18148755	38	1.091e-04	2.555e-02	BIN1:272 DSG2:400 KCNQ1:423 CAV1:468 CAMK2D:639 DSC2:1099
Non-Motile Cilium Assembly (GO:1905515)	0.19944967	31	1.221e-04	2.741e-02	ARL13B:55 CEP89:262 IFT122:403 C2ND:373 TOGARARI:539 TBC1D32:627
Actin-Myosin Filament Sliding (GO:00327	-0.33279429	11	1.326e-04	2.804e-02	MYH2:100 MYH8:261 MYH7:580 TNNT2:472 MYL1:1074 TPM1:1112
Alpha-Amino Acid Catabolic Process (GO:1	0.20476280	29	1.361e-04	2.804e-02	TAT:217 SARDH:303 ALDH4A1:339 KMO:340 HIBCH:752 HMGCL1:1047
Regulation Of Chromosome Segregation (GO	0.22939119	23	1.405e-04	2.804e-02	NCAPD2:110 SMC4:118 SMC2:883 CDCA2:983 NUMA1:1002 NCAPG:1278
Anterograde Trans-Synaptic Signaling (GO	-0.08117521	181	1.730e-04	3.214e-02	NPTX1:32 GABRR1:52 RPS6KA2:67 NTSR1:129 GABRB3:145 SLC18A2:159
Recombinational Repair (GO:0000725)	0.11548165	89	1.697e-04	3.214e-02	RHNO1:112 HELQ:313 MCM9:315 AUNIP:570 ERCC5:638 RAD51AP1:644
Cardiac Conduction (GO:0061337)	-0.16379330	43	2.050e-04	3.563e-02	BIN1:272 DSG2:400 KCNQ1:423 CAV1:468 CAMK2D:639 SLC9A1:900
Sister Chromatid Segregation (GO:0000819	0.18690937	33	2.037e-04	3.563e-02	CHAMP1:73 SMC4:118 SPAG5:334 ESPL1:356 KIFC1:401 KIF18B:468
Monocarboxylic Acid Catabolic Process (GO	0.23780537	20	2.324e-04	3.826e-02	CYP26C1:147 ACOT8:398 HOGA1:1129 LPN3:1469 ACOX1:1805 FAAH:1880
Sensory Perception Of Light Stimulus (GO	-0.10880963	96	2.344e-04	3.826e-02	TRPM1:17 VAX2:19 GNAT2:25 CRYAA:118 CABP2:122 OAT:133
NADH Dehydrogenase Complex Assembly (GO:	0.15189251	48	2.745e-04	4.225e-02	EC5IT:87 NDUFAP2:173 NDUF53:424 TMEM126A:742 NDUF4F1:941 NDUFEB:1065
Mitochondrial Respiratory Chain Complex	0.15189251	48	2.745e-04	4.225e-02	EC5IT:87 NDUFAP2:173 NDUF53:424 TMEM126A:742 NDUF4F1:941 NDUFEB:1065
Positive Regulation Of Chromosome Separa	0.31357597	11	3.170e-04	4.745e-02	NCAPD2:110 SMC4:118 SMC2:883 CDCA2:983 NUMA1:1002 NCAPG:1278
Maturation Of SSU-rRNA (GO:0030490)	0.18127202	32	3.762e-04	5.478e-02	RPS16:467 TSR2:543 RRP36:654 SRFBP1:769 TSRI:953 WDR46:1026
DNA Metabolic Process (GO:0006259)	0.06255285	274	3.894e-04	5.521e-02	PARG:46 MBD4:124 MSH3:225 PARP4:284 MCM9:315 WDHHD1:330
Cardiac Muscle Contraction (GO:0006044)	-0.18594522	30	4.256e-04	5.880e-02	RYR2:386 KCNQ1:423 MYH7:580 CAMK2D:639 TNNT2:672 GRK2:946
Maturation Of SSU-rRNA From Tricrionic	0.20266026	25	4.540e-04	5.966e-02	UTP20:93 RPS16:467 TSR2:543 RRP36:654 KR11:817 TSRI:953

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HAMAL_APOPTOSIS_VIA_TRAIL_UP	0.08804919	591	3.507e-13	2.276e-09	SPG11:2 AGGF1:5 PJA2:7 ENPP2:30 KIF16B:64 CEP192:65
REACTOME_NEURONAL_SYSTEM	-0.09845701	382	4.581e-11	1.486e-07	ADCY1:8 CAMK2B:11 SHANK2:13 GABRR1:52 RPS6KA2:67 DBNL8:81
REACTOME_MUSCLE_CONTRACTION	-0.13868862	185	8.270e-11	1.788e-07	CAMK2B:11 ITPR1:66 KCNJ12:182 NEB:190 MYBPCC2:194 SCN1A:219
NIKOLSKY_BREAST_CANCER_2Q0I2_Q13_AMPLICO	-0.15450244	127	1.892e-09	3.069e-06	NPBW2:28 CABLES2:69 ZBTB46:105 NTSR1:129 MYT1:146 TCF15:147
MARTENS_TRETINOIN_RESPONSE_UP	-0.06750341	675	2.805e-09	3.418e-06	MMEL1:5 SHANK2:13 TRIM29:138 CHST1:523 NPTX1:32 KLK7:35
REACTOME_G_ALPHA_I_SIGNALLING_EVENTS	-0.10554753	267	3.161e-09	3.418e-06	ADCY1:8 CAMK2B:11 GNAT2:25 NPBW2:28 OPN1LW:63 ITPR1:66
FISCHER_DREAM_TARGETS	0.05805156	873	7.771e-09	7.203e-06	MDM1:27 MTBP:32 FANCM:50 ARL13B:55 CEP192:65 FANCE:108
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	-0.09208892	330	9.845e-09	7.984e-06	ADGRB1:12 BNC1:70 VIPR1:74 PRDM16:141 NKX2-4:170 CHRNA4:175
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	-0.05487009	15	4.284e-08	2.779e-05	EP400:1 SFSWAP:9 GOLGA3:231 P2RX2:302 ANKLE2:424 MMP17:427
REACTOME_SIGNALING_BY_GPCR	-0.06452432	626	4.272e-08	2.779e-05	ADCY1:8 CAMK2B:11 TIAM2:14 TACR2:22 GNAT2:25 NPBW2:28
MIKKELSEN_DEVELOPMENTAL_BIOLOGY	-0.06822525	553	4.803e-08	2.833e-05	CAMK2B:11 ADGRB1:12 CDCP1:40 CNTN4:72 NTSR1:129 SLC15A1:144
REACTOME_MEF_FIBROBLAST_BIOLOGY	-0.05329471	909	6.595e-08	3.566e-05	COL6A1:44 L1CAM:57 RPS6KA2:67 KRT80:76 DSG3:88 COL6A2:89
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	-0.25221842	38	7.482e-08	3.734e-05	LFNG:211 INTS1:368 GRIFIN:465 TTYH3:586 ZFAND2A:635 GET4:747
JOHNSTONE_PARVB_TARGETS_3_DN	0.05622156	771	1.344e-07	6.231e-05	MDM1:27 MTBP:32 CCPG1:43 FAM13B:68 PPP4R3B:88 MAN2A1:91
DUTERTE_ESTRADIOL_RESPONSE_24HR_UP	0.08589001	308	2.343e-07	1.013e-04	KCNK15:16 NCAPD2:110 ZWLICH:138 STIL:180 MYO19:201 TET2:202
MARSON_BOUND_BY_E2F4_UNSTIMULATED	0.05943335	65	3.060e-07	1.241e-04	MDM1:27 MTBP:32 TBCD:139 KIAA0825:42 FANCM:50 NCAPD2:110
BLANCO_MELO_BRONCHIAL_EPITHELIAL_CELLS_I	0.11450159	163	4.719e-07	1.713e-04	RHNO1:112 KIF24:175 CLCN3:239 ATAD5:270 IRAG2:285 WDHHD1:330
REACTOME_INFECTIOUS_DISEASE	-0.05343648	779	4.752e-07	1.713e-04	SH3GL3:37 ADCY1:8 ACE2:37 TLR8:38 ITPR1:66 DOCK1:90
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	0.06087136	588	5.275e-07	1.801e-04	AGGF1:5 LEPROTL1:19 CCG1:43 SCD:44 PPP4R3B:88 ITC9:98
DACOSTA_UV_RESPONSE_VIA_ERCC3_COMMON_DN	0.06791195	445	1.001e-06	3.093e-04	PJA2:7 ZNF292:17 MDN1:21 AKAP9:24 AFF1:35 DST:45
NUYTEN_EZH2_TARGETS_DN	0.04783517	930	9.551e-07	3.093e-04	PJA2:7 LEPROTL1:19 DDHD2:23 MTBP:32 WNK4:51 NBR1:69
DACOSTA_UV_RESPONSE_VIA_ERCC3_DN	0.05051916	815	1.150e-06	3.393e-04	SPG11:2 TMEM131L:3 PJA2:7 ZNF292:17 MDN1:21 AKAP9:24
REACTOME_CARDIAC_CONDUCTION	-0.12775290	120	1.372e-06	3.561e-04	CAMK2B:11 ITPR1:66 KCNJ12:182 SCN1A:219 ATP2B3:298 SCN2A:338
WP_GPCRS_CLASS_A_RHODOPSINLIKE	-0.09662510	211	1.372e-06	3.561e-04	NPBW2:28 OPN1LW:63 OPN3:75 NTSR1:129 FFAR3:179 HRH3:271
KAUFMANN_DNA_REPAIR_GENES	0.09706905	210	1.299e-06	3.561e-04	FANCM:50 TEPI:94 FANCE:108 SMC4:118 MBD4:124 MSH3:225
REACTOME_KERATINIZATION	-0.14989450	85	1.780e-06	4.441e-04	KRT80:76 DSG3:88 KRT13:96 TGM1:109 TGM5:140 KRT5:238
MIKKELSEN_NPC_HCP_WITH_H3K27ME3	-0.07669899	329	1.859e-06	4.469e-04	ADGRB1:12 BNC1:70 VIPR1:74 PRDM16:141 CHRNA4:175 CADM3:176
WINTER_HYPOXIA_METAGENE	-0.00239098	222	2.195e-06	5.087e-04	IGFBP1:41 IGFBP3:43 TGM2:150 SLC6A16:152 NTSR1:129 SORL1:198
REACTOME_GPCR_LIGAND_BINDING	-0.06925642	398	2.304e-06	5.154e-04	TACR2:22 NPBW2:28 OPN1LW:63 VIPR1:74 OPN3:75 UTS2R:91
BENPORATH_ES_WITH_H3K27ME3	-0.04464541	978	2.830e-06	6.120e-04	PLEC:10 CAMK2B:11 VAX2:19 NKPD1:21 FRMPD4:31 NPTX1:32
REACTOME_TRANSMISSION_ACROSS_CHEMICAL_SY	-0.08698466	243	3.156e-06	6.605e-04	ADCY1:8 CAMK2B:11 GABRR1:52 RPS6KA2:67 GLUL:108 GABRB3:145
GOBERT_OLIGODENDROCYTE_DIFFERENTIATION_I	0.05855762	542	3.487e-06	7.070e-04	FBN2:31 CEP192:65 NCAPD2:110 SMC4:118 ZWLICH:138 HAUS6:171
DAZARD_RESPONSE_TO_UV_NHEK_DN	0.08012816	281	4.011e-06	7.885e-04	TMEM131L:3 AKAP9:24 AFF1:35 DST:45 SMC4:118 SCAF11:166
WP_STRIATED_MUSCLE_CONTRACTION_PATHWAY	-0.23368138	32	4.777e-06	8.854e-04	NEB:190 MYBPCC2:194 MYH8:261 TNNT3:391 DES:561 TNNT2:672
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	-0.08253265	261	4.649e-06	8.854e-04	TACR2:22 NPBW2:28 GABRR1:52 VIPR1:74 UTS2R:91 NTSR1:129
REACTOME_NEUROTRANSMITTER_RECEPTORS_AND	-0.09791101	182	5.394e-06	9.457e-04	ADCY1:8 CAMK2B:11 GABRR1:52 RPS6KA2:67 GABRB3:145 CHRNA4:175
REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE	-0.14887774	78	5.539e-06	9.457e-04	KRT80:76 DSG3:88 KRT13:96 TGM1:109 TGM5:140 KRT5:238
RODRIGUES_THYROID_CARCINOMA_ANAPLASTIC_U	0.05352207	627	5.405e-06	9.457e-04	WDR36:74 PPP4R3B:88 ITC9:98 SMC4:118 ZWLICH:138 HAUS6:171
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	-0.05395310	607	6.352e-06	1.057e-03	CAMK2B:11 TRIM29:18 IL37:39 TNFRSF4:46 RPS6KA2:67 P4HB:68
RICKMAN_METASTASIS_DN	-0.08647918	227	7.447e-06	1.208e-03	TRIM29:18 ITGB4:26 CDCP1:40 RNH1:49 DSG3:88 IL18:107

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HIV Infections	-0.06391070	626	6.598e-08	6.473e-04	PLEC:10 TLR8:38 VIPR1:74 APOA1:84 ISG20:101 IL18:107
Primary microcephaly	0.12933094	104	5.362e-06	2.630e-02	FANCM:50 FANCE:108 STIL:180 CDK5RAP2:200 TRMT10A:259 MFS2DA:265
Congenital cerebral hernia	0.18408151	46	1.582e-05	5.172e-02	CRY1:287 WDPCP:289 RDH12:387 ARHGAP31:476 KCNJ13:897 FLNB:927
CATARACT_AUTOSOMAL_DOMINANT	-0.37959311	10	3.232e-05	5.285e-02	CRYAA2:47 GJA8:54 CRYAA:118 CRYBB2:221 BFSP2:447 MIP:466
Nuclear cataract	-0.21448307	32	2.698e-05	5.285e-02	HSF4:36 GJA8:54 CRYAA:118 TGM2:150 CRYBB2:221 FTO:380
Virus Diseases	-0.04780109	678	2.721e-05	5.285e-02	ADGRB1:12 LEXM:29 ACE2:37 TLR8:38 TNFRSF4:46 OPN1LW:63
Mouth Neoplasms	-0.11577652	106	3.925e-05	5.500e-02	UPP1:15 HMGCS2:50 IL18:107 PTK2:165 EGFR:212 CAV1:468
Autism Spectrum Disorders	-0.05362741	481	6.421e-05	7.874e-02	SFSWAP:9 SHANK2:13 L1CAM:57 HERC2:58 JAKMIP1:60 ITPR1:66
Cataract, Central Saccular, With Sutral	-0.46255736	6	8.714e-05	7.939e-02	GJA8:54 CRYBB2:221 BFSP2:447 MIP:466 CRYBA1:777 CRYGS:1458
Night blindness, congenital stationary	-0.22853271	25	7.677e-05	7.939e-02	TRPM1:17 PDE6B:48 CACNA2D4:167 GRK7:178 NYX:196 GRM6:268
Asthma	-0.03632998	1057	8.902e-05	7.939e-02	MMEL1:5 SFSWAP:9 ISGF1:16 TACR2:22 ITGB4:26 TLR8:38
Complete congenital stationary night bli	-0.30084589	14	9.734e-05	7.958e-02	TRPM1:17 PDE6B:48 NYX:196 GRM6:268 FGFR2:645 CACNA1F:1166
Anemia, hereditary spherocytic hemolytic	0.49056707	5	1.453e-04	8.027e-02	SPTA1:47 ANK1:53 SLC4A1:195 EPB42:235 SPTB:244 NA
Cardiomyopathies	-0.05470512	420	1.339e-04	8.027e-02	PLEC:10 ACE2:37 UTS2R:91 MYH2:100 ISG20:101 IL18:107
Nasopharyngeal carcinoma	0.04593978	603	1.365e-04	8.027e-02	IGFBP3:43 OPN1LW:63 TRAF2:80 ISG20:101 IL18:107 CD40:110
Oguchi disease	-0.27810641	16	1.176e-04	8.027e-02	TRPM1:17 PDE6B:48 GRK7:178 NYX:196 GRM6:268 FGFR2:645
Sloping forehead	0.12366245	79	1.473e-04	8.027e-02	STIL:180 CDK5RAP2:200 MFS2DA:265 WDPCP:289 ERCC6:296 ATR:347
Spherocytosis	0.49056707	5	1.453e-04	8.027e-02	SPTA1:47 ANK1:53 SLC4A1:195 EPB42:235 SPTB:244 NA
Lymphoma	-0.03478731	1065	1.664e-04	8.592e-02	ADGRG7:4 OPN1LW:63 TRAF2:80 ISG20:101 IL18:107 CD40:110
NIGHT BLINDNESS, CONGENITAL STATIONARY,	-0.30860985	12	2.144e-04	8.795e-02	TRPM1:17 PDE6B:48 NYX:196 GRM6:268 CACNA1F:1166 CABPA:1637
Arthritis	-0.04754330	516	2.476e-04	8.795e-02	TLR8:38 IL37:39 IGFBP3:43 TRAF2:80 APOA1:84 CLEC2A:93
Ciliopathies	0.08160786	170	2.510e-04	8.795e-02	ARL13B:55 INVS:106 PKHD1:223 WDPCP:289 NPHP3:287 ATR:347
Epidermolysis Bullosa Simplex	-0.23342433	21	2.136e-04	8.795e-02	PLEC:10 ITGB4:26 KRT80:76 TGM5:140 KRT5: